Economics 351X Test 1 Fall 2026 Prof. Hill Test Version B

Instructions: Please fill out your name on this page and sign the honesty pledge. On the Akindi form, fill in your name, student ID, and exam version. Once you've worked out your answers, transfer them to the Akindi answer sheet. You have 75 min – use your time accordingly.

Name: _____

I pledge that I have neither given nor received aid on this examination.

Signature: _____

Chapter 2

Derivative Rules

General Function	Derivative	Examples
$f(x) = c + ax^b$	$\frac{df(x)}{dx} = abx^{(b-1)}$	$f(x) = 3 + 5x^2 \Rightarrow \frac{df(x)}{dx} = 10x$
$f(x_1, x_2) = ax_1^b + cx_2^d$	$\frac{\partial f(x_1,x_2)}{\partial x_1} = abx_1^{(b-1)}$ $\frac{\partial f(x_1,x_2)}{\partial x_2} = cdx_2^{(d-1)}$	$f(x) = 5x_1^2 + 3x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_1} = 10x_1$ $f(x) = 5x_1^2 + 3x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_2} = 3$
$f(x_1, x_2) = ax_1^b x_2^c$	$\frac{\partial f(x_1,x_2)}{\partial x_1} = abx_1^{(b-1)}x_2^c$ $\frac{\partial f(x_1,x_2)}{\partial x_2} = acx_1^b x_2^{(c-1)}$	$f(x) = 5x_1^2x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_1} = 10x_1x_2$ $f(x) = 5x_1^2x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_2} = 5x_1^2$
Chain rule	$h(x) = f(g(x))$ $h'(x) = f'(g(x))g'(x)$	$h(x) = (7 + 3x^2)^5$ $h'(x) = 5(7 + 3x^2)^4 \times (6x)$
Product rule	$h(x) = f(x)g(x)$ $h'(x)=f'(x)g(x)+f(x)g'(x)$	$h(x)=(7 + 3x^2)(6 + 5x^3)$ $h'(x)=(6x)(6+5x^3)+(7+3x^2)(15x^2)$

Price Elasticity of Supply

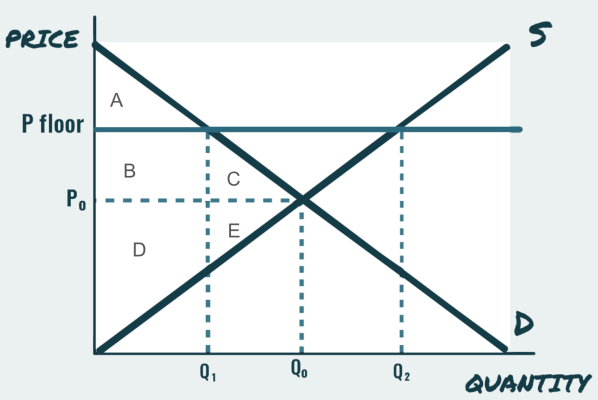
$$E_{Q^S,p} = \left(\frac{dQ^S(p)}{dp}\right)\frac{p}{Q^S(p)}.$$

Chapter 4

Price Elasticity of Demand

$$E_{Q_1^*,p_1} = \left(\frac{\partial Q_1^*(p_1,p_2,I)}{\partial p_1}\right)\frac{p_1}{Q_1^*(p_1,p_2,I)}.$$

Or E= percent change in Q / percent change in P



1. Refer to the above figure, the government implements a price floor above the equilibrium. The elasticity of demand is -2, the government agrees to buy up any excess supply, what happens to producer's revenue?

a) It stays the same
b) It increases
c) It decreases
d) Need more information

2. Imagine that demand is Q^D=105-5P and supply is Q^S=10P. The government institutes a price ceiling at P=$5. What will be the result?

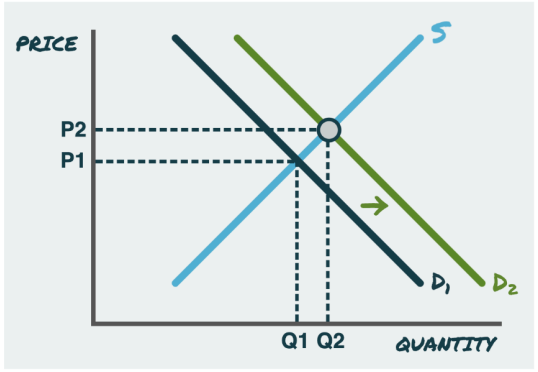
a) A surplus of 30 units
b) A shortage of 30 units
c) A shortage of 20 units
d) A surplus of 20 units
e) There will not be a shortage, nor a surplus

3. Imagine the government would like to decrease alcohol consumption by 50%. They know that the demand for alcohol has an elasticity of -0.2. By how much should they increase prices by? Would you expect the drop in alcohol consumption to increase, decrease, or stay the same in the long run?

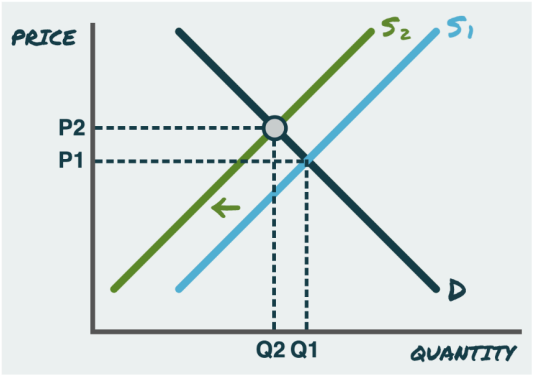
a) 250%, increase
b) 50%, decrease
c) 25%, decrease
d) 10%, increase
e) None of the above

Use the following answer set for the next 3 questions

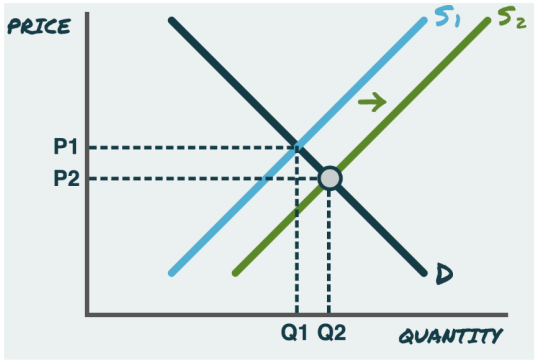
A)



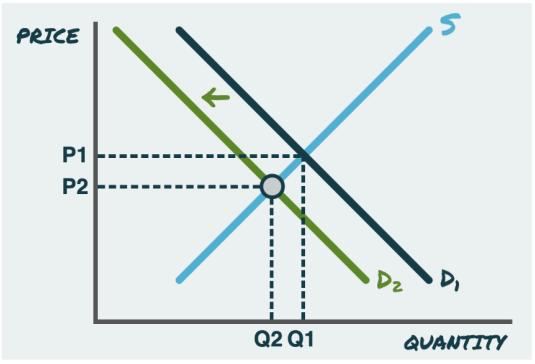
B)



C)



D)



4. The price of socks rise, which graph best depicts what will happen in the market for shoes (a complement to socks)?

a) graph A b) graph B c) graph C **d) graph D**

5. The price of beef rises significantly. What graph represents what will happen to the Market for hamburgers (beef is an ingredient to Hamburgers)?

a) graph A **b) graph B** c) graph C d) graph D

6. Incomes rise. What graph represents what will happen to the market for inferior goods?

a) graph A b) graph B c) graph C **d) graph D**

7. The demand curve is given by the equation Q=30-2P, the supply curve is given by the equation Q=P. What is consumer surplus at the equilibrium price?

a) 50
b) 75
c) 100
d) 125
e) None of the above

8. Which of the following would shift the supply curve to the left in the market for oranges? (MARK ALL THAT APPLY)

a) A disease destroys orange trees
b) The price of farm labor increases
c) The price of apples (a substitute for oranges) falls
d) A new technology is invented that increases orange tree yields
e) Incomes rise

9. What is NOT an example of unintended consequences?

A. A bounty of snakes leading to people raising snakes
B. A 3 point percentage bonus leading to a player not shooting threes
C. A price control designed to make food cheaper making food scarce
D. All of the above are examples of unintended consequences

10. Bicycles were–

a) Celebrated when they were first invented
b) Caused a moral panic when they were first invented
c) Covered in the moonshine stimulus podcast
d) An example of perfectly inelastic supply

11. Jack buys a $5 coffee from his local coffee shop. It is usually $4, but the owner has recently raised the price. Assume the market is at a perfectly competitive equilibrium. Which of the following statements is true (MARK ALL THAT APPLY)?

a) Total surplus would have been higher at a different price
b) Jack's consumer surplus is negative
c) Producer surplus is negative
d) Consumer surplus has decreased
e) None of the above

12. The examples of the glass maker from Rome and the engineer from Kodak illustrate what idea

a) Innovators are often rewarded for their new inventions
b) Markets decrease consumer surplus
c) The elasticity of demand for new products is less than one in absolute terms
d) It is hard to predict where innovation will occur

Suppose the market demand curve is

$Q_d=50-2P$

and the market supply curve is

$Q_s=10+3P$

13. What is the equilibrium in this market in perfect competition?

a) P=6, Q=38
b) P=8, Q=34
c) P=10, Q=30
d) P=12, Q=26
e) none of the above

14. Imagine the government would like to increase EV purchases by 20%. They know that the demand for EVs has an elasticity of -4. By how much should they decrease prices by?

a) 5%
b) 80%
c) 5 dollars
d) 80 dollars
e) None of the above

15. Peanut Butter becomes cheaper. At the same time, strawberries (an input into Jam) become less expensive. What do you expect to happen to the price and quantity sold of Jam (a complement to Peanut Butter) in the Jam Market?

a) Price increases, quantity change is indeterminate
b) Price change is indeterminate, quantity increases
c) Price change is indeterminate, quantity decreases
d) Price decreases, quantity decreases
e) Price decreases, quantity change is indeterminate

16. The demand curve is given by the equation Q=30-2P, the supply curve is given by the equation Q=P. What is consumer surplus at the equilibrium price?

a) 50
b) 75
c) 100
d) 125
e) None of the above

17. Which of the following goods would have the least elastic demand (pick the best answer)?

a) Sunscreen at the beach
b) Luxury Cars
c) Vacations
d) A fancy treadmill

18. What is a possible second order effect of price ceilings (mark all that apply)?

a) The decline in quality of a good
b) Underground markets
c) Discrimination
d) Bribery
e) Long lines

19. The demand for apples has an elasticity of demand of -5. Which of the following would increase revenue in the market apples? (MARK ALL THAT APPLY)

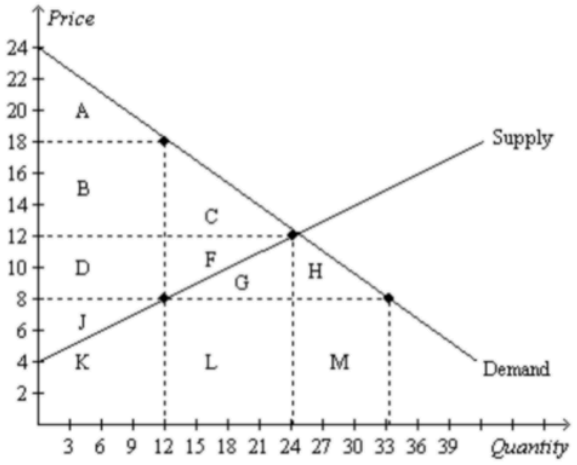
a) A price ceiling below the market equilibrium
b) A price floor above the market equilibrium
c) An increase in price of oranges (a substitute for apples)
d) An increase in price of water (an input into apple production)
e) A new technology that enables cheaper production of apples

20. Imagine that demand is Q^D=105-5P and supply is Q^S=10P, what is the deadweight loss if a price ceiling is implemented at $3 compared to perfect competition?

a) 0
b) 30
c) 60
d) 240
e) None of the above

21. A person's demand for an iphone charger would be _____ elastic when their battery life is at 1% vs. 99%?

a) More
b) Less
c) The same



22. Refer to the above graph. What would the deadweight loss be if a price ceiling of $8 was imposed on the market?

a) CF

b) ABD

c) DFJ

d) C

e) None of the above

EXAM will be Counted out of 22, the following three questions are bonus questions, 2 from the readings, one just extra credit

23. Who won the bet in the video about Paul Ehrlich?
    a) Paul Ehrlich
    b) Julian Simon
    c) It was a tie
    d) Simon in the short run, but Ehrlich in the long run

24. San Francisco offers a nice experiment in Rent Control, what were some of the consequences of Rent Control in that city? (MARK ALL THAT APPLY)
    a) An increase in market rents
    b) Conversion of apartments into owner-occupied housing
    c) Potential tenants taking out ads in the newspaper looking for apartments
    d) Severe shortages after the 1906 earthquake

25. When there is 10 minutes left I will reveal the answer to this question
    a) A
    b) B
    c) C
    d) The opportunity cost of waiting it too high, I am leaving.

D also accepted